Chapter 4: Synthetic Data Generation

4.1 What is synthetic data ?

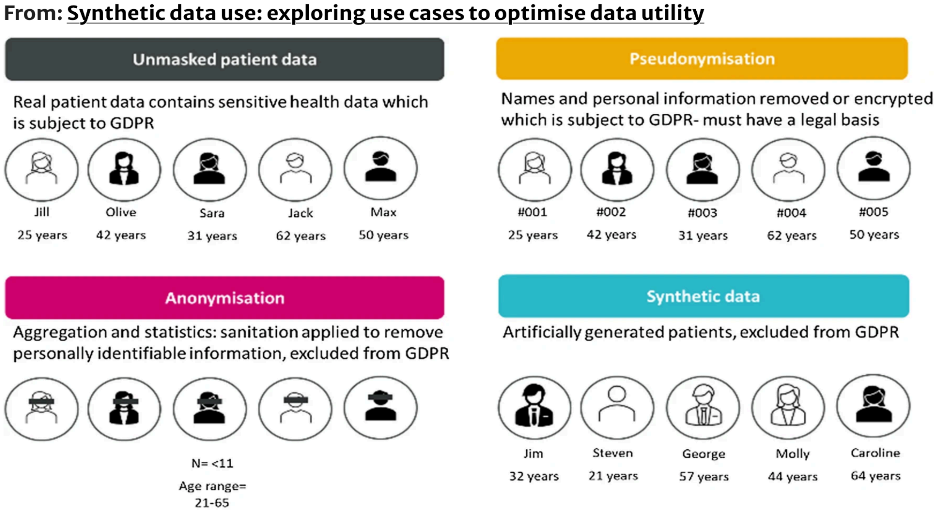
Synthetic data are increasingly being recognized for their potential to address serious real-world challenges in various domains. They provide innovative solutions to combat the data scarcity, privacy concerns, and algorithmic biases commonly used in machine learning applications. Synthetic data preserve all underlying patterns and behaviors of the original dataset while altering the actual content.

4.2 Why is it important ?

Importance	What synthetic data provides
1. Privacy protection	Reduces the need to directly share or process real individual-level data
2. Data accessibility	Allows researchers to explore and work with data when real data access is restricted
3. Research and ML development	Supports model development, testing, comparison and data augmentation
4. Controlled experimentation	Allows researchers to control sample size and data-generating mechanisms
5. Data sharing and collaboration	Facilitates teaching, software testing, collaboration and external sharing

4.3 Difference between real, anonymized and Synthetic data:

From: Synthetic data use: exploring use cases to optimise data utility



The different privacy enhancing techniques that can be used

4.4 Synthetic data Generation methods:

Synthetic data generation involves fitting a statistical or computational model to the original dataset and subsequently sampling new observations from the fitted model. The objective is to reproduce relevant statistical characteristics and dependencies of the original data while generating records that are not direct copies of the original observations.

D real→Fit generative model→Learn distributions/dependencies→D synthetic

The method is stochastic, as it involves repeated generation from the fitted model, which creates different synthetic data .
4.4.1 Parametric synthetic data generation:

Concept: Fit statistical models to the observed data, estimate their parameters, and then generate new synthetic observations by sampling from the fitted distributions.

where:

- Yobs = observed data
- X = predictors/covariates
- $f(\cdot)$ = assumed probability distribution/model
- θ = parameters estimated from the observed data

The synthesiser first **fits the model to the observed data** and estimates θ. Then synthetic observations are generated from the fitted model.

4.4.1.2 The most important concept: conditional synthesis

The paper explains that synthpop generally does **not model the entire joint distribution in one step**.

Instead, it decomposes the joint distribution into a sequence of conditional distributions.

For example, suppose you have:

X1,X2,X3,X4

The synthesis could proceed as:

$P(X1,X2,X3,X4)=P(X1)P(X2\mid X1)P(X3\mid X1,X2)P(X4\mid X1,X2,X3)$

So:

1. Generate X1
2. Generate $X2\mid X1$
3. Generate $X3\mid X1,X2$
4. Generate $X4\mid X1,X2,X3$

This is a **very important point for your handbook**.

It means that parametric synthetic data generation can preserve relationships between variables by modelling **conditional relationships**.

4.4.1.3 What parametric models does synthpop use?

The paper gives you an excellent table that you can use to explain this.

Variable type	**Parametric method**	**Model**
Numeric	norm	Normal linear regression
Numeric	normrank	Normal regression preserving marginal distribution
Numeric	pmm	Predictive mean matching
Binary	logreg	Logistic regression
Categorical (>2 levels)	polyreg	Polytomous logistic regression
Ordered categorical	polr	Ordered logistic regression

So, for example: **Continuous variable**

Suppose: $Y = \text{age}$ and you have predictors such as sex and education.

A parametric model could be: $\text{Age} = \beta_0 + \beta_1 \text{Sex} + \beta_2 \text{Education} + \epsilon$

The parameters β are estimated from the observed data.

Synthetic age values are then generated using the fitted model.

Binary variable

Suppose: $Y = \text{disease status}$, where $Y=1$ means disease and $Y=0$ means no disease.

A logistic regression model could be: $\text{logit}\{P(Y=1 \mid X)\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots$

After estimating the β's from the real data, probabilities are calculated for synthetic individuals and new binary outcomes are sampled.

Categorical variable

For a variable such as education with several categories, synthpop can use **polytomous logistic regression**.

For an ordered variable, it can use **ordered logistic regression**.

4.4.1.4 The synthesis sequence is important

This is another point worth taking from the paper.

Suppose your dataset contains:

- age
- sex
- education
- income
- disease
- treatment
- outcome

You might specify:

Age→Sex→Education→Income→Disease→Treatment→Outcome

The model for each variable can use variables that have already been synthesised.

For example:

$P(\text{Income} \mid \text{Age}, \text{Sex}, \text{Education})$

then:

$P(\text{Disease} \mid \text{Age}, \text{Sex}, \text{Education}, \text{Income})$

and so on.

This is controlled in synthpop using:

- visit.sequence
- predictor.matrix

This is particularly relevant to **your causal-inference work**, because the ordering and predictor selection determine which relationships are reproduced in the synthetic data.

4.5**Non parametric - Machine Learning Methods:**

Parametric methods require assumptions about the functional form and distribution of the data.
Real-world datasets may contain **nonlinear relationships, interactions, thresholds, and complex dependencies**.
Machine-learning methods can learn these relationships directly from the observed data.
ML-based synthesis therefore provides a more flexible alternative to traditional parametric synthesis.

4.5.1**Classification and Regression Trees (CART)**

Include:

- CART is a **non-parametric tree-based method** used for synthetic data generation.
- It was introduced for synthetic data generation by **Reiter (2005)**.
- CART can handle:
  - continuous variables,
  - binary variables,
  - categorical variables.
- It recursively partitions observations into relatively homogeneous groups.
- The resulting terminal nodes represent groups with similar characteristics.
- Synthetic observations are generated by sampling from the distributions represented by the relevant terminal nodes.
- CART can capture:
  - nonlinear relationships,
  - interactions,
  - threshold effects.
- Unlike conventional regression models, CART does not require a predefined linear functional form.

Key equation/concept

For a variable Yj:

$P(Yj \mid X1,\ldots,Xk)$

is estimated using the tree structure, and synthetic values are sampled from the resulting conditional distribution.

4.5.2 **Conditional Inference Trees (CTREE)**

Include:

- CTREE is a **conditional inference tree** approach.
- It provides an alternative to conventional CART.
- CTREE uses **statistical hypothesis tests** to determine whether a predictor is associated with the response and whether a split should be performed.
- The tree therefore grows based on statistically detected associations rather than solely on impurity reduction.
- CTREE can accommodate different types of variables.
- It can capture nonlinear associations and interactions.
- CTREE is available as a synthesis method in synthpop.

A simple conceptual representation:

$H0{:}Xj \perp Y$, If sufficient evidence exists against H0, the algorithm considers splitting the node according to Xj.

4.5.3 Random forest

Random Forest is an **ensemble machine-learning method** consisting of many decision trees.

- Instead of relying on a single tree, RF builds multiple trees using:
  - **Bootstrap samples** of the original data.
  - Randomly selected subsets of predictors at each split.

- The predictions from the individual trees are combined to obtain a more stable prediction.

How Random Forest can be used for synthetic data

For synthetic data generation, RF can be used to model the relationship between variables and then generate new observations.

The basic procedure is:

Original data → Fit Random Forest models → Estimate conditional distributions → Generate synthetic observations

For example, if the variables are:

- $X1,X2,X3$: predictors
- Y: variable to be synthesised

RF can model:

$Y=f(X1,X2,X3)+\epsilon$

For a categorical Y, RF estimates the probability of different categories.
For a continuous Y, RF can be used to estimate the conditional distribution of Y given the predictors.

The generated value is then sampled from this estimated distribution rather than simply taking the RF prediction.

4.6 Comparison for Ctree, CART, RF

Feature	CART	CTREE	Random Forest (RF)
Full name	Classification and Regression Trees	Conditional Inference Trees	Random Forest
Basic approach	Single decision tree	Single statistically guided decision tree	Ensemble of many decision trees
Main principle	Recursive binary splitting	Statistical hypothesis testing determines splits	Bootstrap samples + random predictor selection + multiple trees
Number of trees	One	One	Many
Variable selection	Selects split giving the best impurity reduction	Uses statistical tests to select significant associations	Random subset of predictors considered at each split

Split criterion	Gini impurity, entropy, residual sum of squares, etc.	Statistical permutation/hypothesis tests	Usually impurity-based criteria within individual trees
Non-linear relationships	✓	✓	✓✓
Interactions	✓	✓	✓✓
Mixed variable types	✓	✓	✓
Overfitting tendency	Relatively high	Generally lower than CART	Lower due to ensemble averaging
Prediction stability	Lower	Moderate	High
Interpretability	High	High	Lower
Computational complexity	Low	Moderate	Higher
Main advantage	Simple and interpretable	Reduces some biases of conventional tree splitting	Strong predictive performance and stability
Main limitation	Can be unstable and overfit	More computationally involved than simple CART	Less interpretable and computationally intensive
Synthetic-data application	Models conditional distributions using a tree-based structure	Models conditional distributions using statistically selected splits	Uses an ensemble to model complex conditional relationships
Ability to capture complex patterns	Moderate	Moderate–high	High
Suitable for small datasets	Can be used, but trees may be unstable	Can be useful, but statistical tests may have limited power	Can perform poorly if the sample is very small
Synthetic-data generation	Can generate observations from terminal-node distributions	Can generate observations from terminal-node distributions	Requires a mechanism to sample from estimated conditional distributions

Typical role in synthesis	Basic tree-based synthesis	Statistically controlled tree-based synthesis	More flexible ensemble-based synthesis

4.7 Probabilistic Synthetic Data Generation

4.7.1 Introduction

- Probabilistic synthetic data generation models the **joint probability distribution** of the variables in the original dataset.
- Synthetic observations are then generated by **sampling from the estimated probability distribution**.
- Bayesian networks provide a graphical framework for representing dependencies among variables.
- Instead of modelling the complete joint distribution directly, Bayesian networks decompose it into a set of **local conditional probability distributions**.

4.7.2 Bayesian Networks

- A Bayesian network is a **probabilistic graphical model** represented by a **directed acyclic graph (DAG)**.
- **Nodes** represent variables.
- **Directed edges** represent probabilistic dependencies between variables.
- Each variable is conditionally dependent on its **parent variables**.
- The joint distribution can be factorized as:

$$P(X1,\ldots,Xp)=i=1\prod pP(Xi\mid Pa(Xi))$$

where Pa(Xi) denotes the set of parent variables of Xi.

- This factorization allows complex, high-dimensional relationships to be represented through simpler local distributions.

4.7.3 Conditional Independence

- Bayesian networks use **conditional independence** to simplify the representation of the joint distribution.
- The graphical structure describes which variables are conditionally independent given other variables.
- **d-separation** provides a graphical criterion for determining conditional independence in a DAG.
- This reduces the number of relationships that need to be explicitly modelled.

4.7.4 Learning the Bayesian Network

Learning a Bayesian network generally involves two major steps:

1. Structure learning

- Determines which variables are connected.
- Determines the possible direction of relationships.
- Produces the DAG structure.

2. Parameter learning

- Estimates the conditional probability distributions associated with the learned structure.
- For discrete variables, these may be represented using **conditional probability tables**.
- For continuous variables, appropriate continuous distributions can be estimated.

Thus:

Observed data→Structure learning→Parameter learning→Bayesian network

4.7.5 Approaches to Structure Learning

Two major approaches are described by Scutari (2010):

Approach	Description
Constraint-based	Uses conditional-independence tests to identify relationships between variables
Score-based	Searches through candidate network structures and selects the structure that optimizes a statistical score

Examples include:

- **Constraint-based:** Grow-Shrink, IAMB, Fast-IAMB, Inter-IAMB and MMPC.
- **Score-based:** Hill-Climbing.

4.7.6 Generation of Synthetic Data

Once the Bayesian network has been learned:

1. Learn the network structure from the original data.
2. Estimate the parameters of the local conditional distributions.
3. Construct the joint probability distribution.
4. Sample new observations from the learned network.
5. The resulting observations constitute the **synthetic dataset**.

The process can be represented as:

$D_{real} \rightarrow G \rightarrow \theta \rightarrow P(X_1, \ldots, X_p) \rightarrow D_{syn}$

where:

- D_{real} = original dataset
- G = learned network structure
- θ = estimated parameters
- D_{syn} = synthetic dataset

4.7.7 Types of Bayesian Networks

Scutari (2010) discusses two important settings:

- **Discrete Bayesian networks:** variables are represented using multinomial distributions and conditional probability tables.
- **Gaussian Bayesian networks:** continuous variables are modelled using Gaussian distributions and linear relationships.

This distinction is important when applying Bayesian networks to health datasets containing **binary, categorical and continuous variables**.

4.7.8 Incorporating Prior Knowledge

- Bayesian-network structure learning does not necessarily have to rely entirely on the observed data.
- **Domain knowledge** can be incorporated into the network.
- In `bnlearn`, this can be done using:
    - **Whitelist:** relationships that should be present.
    - **Blacklist:** relationships that should not be present.
- This can be particularly useful in health and epidemiological applications where some relationships are already supported by substantive knowledge.

4.7.9 Bayesian Networks and Causal Inference

This distinction should definitely be included in your handbook:

> **A Bayesian network represents probabilistic dependencies; it should not automatically be interpreted as a causal model.**

A directed edge learned from observational data does not by itself establish a causal effect.

For example:

$A \rightarrow Y$

in a learned Bayesian network indicates a dependency represented by the model, but it does **not automatically mean that intervening on A will cause a change in Y**.

This distinction is particularly important when evaluating **causal utility** of synthetic data.

4.7.10 Advantages

- Represents complex dependencies between variables.
- Captures multivariable relationships.
- Provides an interpretable graphical representation.
- Can model high-dimensional data through local conditional distributions.
- Can incorporate prior/domain knowledge.
- Can generate synthetic observations from the learned probability model.

- Supports both discrete and continuous data under appropriate model assumptions.

4.7.11 Limitations

- Structure learning can become computationally challenging as the number of variables increases.
- The learned structure can be sensitive to **sample size**.
- Sparse data can make conditional-independence testing unreliable.
- Different structure-learning algorithms may produce different networks.
- Distributional assumptions may affect parameter estimation and synthetic-data quality.
- A probabilistic relationship should not automatically be interpreted as a causal relationship.

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